

Titanic_Data_Preprocessing

September 10, 2026

```
[1]: import pandas as pd

df = pd.read_csv("Titanic.csv")

print(df.head())
```

```
-----
ModuleNotFoundError                                Traceback (most recent call last)
Cell In[1], line 1
----> 1 import pandas as pd
      3 df = pd.read_csv("Titanic.csv")
      5 print(df.head())

ModuleNotFoundError: No module named 'pandas'
```

```
[2]: import pandas as pd

print(pd.__version__)
```

```
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```
[1]: import pandas as pd

print(pd.__version__)
```

3.0.2

```
[2]: df = pd.read_csv("Titanic.csv")

print(df.head())
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
[3]: print("Shape:", df.shape)

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())
```

Shape: (891, 12)

Data Types:

```
PassengerId    int64
Survived       int64
Pclass         int64
Name           str
Sex            str
Age           float64
SibSp          int64
Parch          int64
Ticket         str
Fare           float64
Cabin          str
Embarked       str
dtype: object
```

Missing Values:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Duplicate Rows: 0

```
[4]: print(df.isnull().sum())
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
[5]: df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

print(df.isnull().sum())
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687

Embarked 0
dtype: int64

```
[6]: from sklearn.preprocessing import StandardScaler

num_cols = ["Pclass", "Age", "SibSp", "Parch", "Fare"]

scaler = StandardScaler()

df[num_cols] = scaler.fit_transform(df[num_cols])

print(df[num_cols].head())
```

	Pclass	Age	SibSp	Parch	Fare
0	0.827377	-0.565736	0.432793	-0.473674	-0.502445
1	-1.566107	0.663861	0.432793	-0.473674	0.786845
2	0.827377	-0.258337	-0.474545	-0.473674	-0.488854
3	-1.566107	0.433312	0.432793	-0.473674	0.420730
4	0.827377	0.433312	-0.474545	-0.473674	-0.486337

```
[7]: from sklearn.preprocessing import OneHotEncoder

cat_cols = ["Sex", "Embarked"]

encoder = OneHotEncoder(sparse_output=False, handle_unknown="ignore")

encoded = encoder.fit_transform(df[cat_cols])

encoded_df = pd.DataFrame(
    encoded,
    columns=encoder.get_feature_names_out(cat_cols),
    index=df.index
)

print(encoded_df.head())
```

	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	0.0	1.0	0.0	0.0	1.0
1	1.0	0.0	1.0	0.0	0.0
2	1.0	0.0	0.0	0.0	1.0
3	1.0	0.0	0.0	0.0	1.0
4	0.0	1.0	0.0	0.0	1.0

```
[8]: print("Statistical Analysis:")
print(df[["Pclass", "Age", "SibSp", "Parch", "Fare"]].describe())
```

Statistical Analysis:

	Pclass	Age	SibSp	Parch	Fare
count	8.910000e+02	8.910000e+02	8.910000e+02	8.910000e+02	8.910000e+02

```

mean    -8.772133e-17  2.272780e-16  4.386066e-17  5.382900e-17  3.987333e-18
std      1.000562e+00  1.000562e+00  1.000562e+00  1.000562e+00  1.000562e+00
min     -1.566107e+00 -2.224156e+00 -4.745452e-01 -4.736736e-01 -6.484217e-01
25%     -3.693648e-01 -5.657365e-01 -4.745452e-01 -4.736736e-01 -4.891482e-01
50%      8.273772e-01 -1.046374e-01 -4.745452e-01 -4.736736e-01 -3.573909e-01
75%      8.273772e-01  4.333115e-01  4.327934e-01 -4.736736e-01 -2.424635e-02
max      8.273772e-01  3.891554e+00  6.784163e+00  6.974147e+00  9.667167e+00

```

```

[9]: print("\nSurvival Count:")
     print(df["Survived"].value_counts())

```

```

Survival Count:
Survived
0      549
1      342
Name: count, dtype: int64

```

```

[10]: print("\nSex Count:")
      print(df["Sex"].value_counts())

      print("\nEmbarked Count:")
      print(df["Embarked"].value_counts())

```

```

Sex Count:
Sex
male      577
female    314
Name: count, dtype: int64

```

```

Embarked Count:
Embarked
S      646
C      168
Q       77
Name: count, dtype: int64

```

```

[11]: import matplotlib.pyplot as plt
      import seaborn as sns

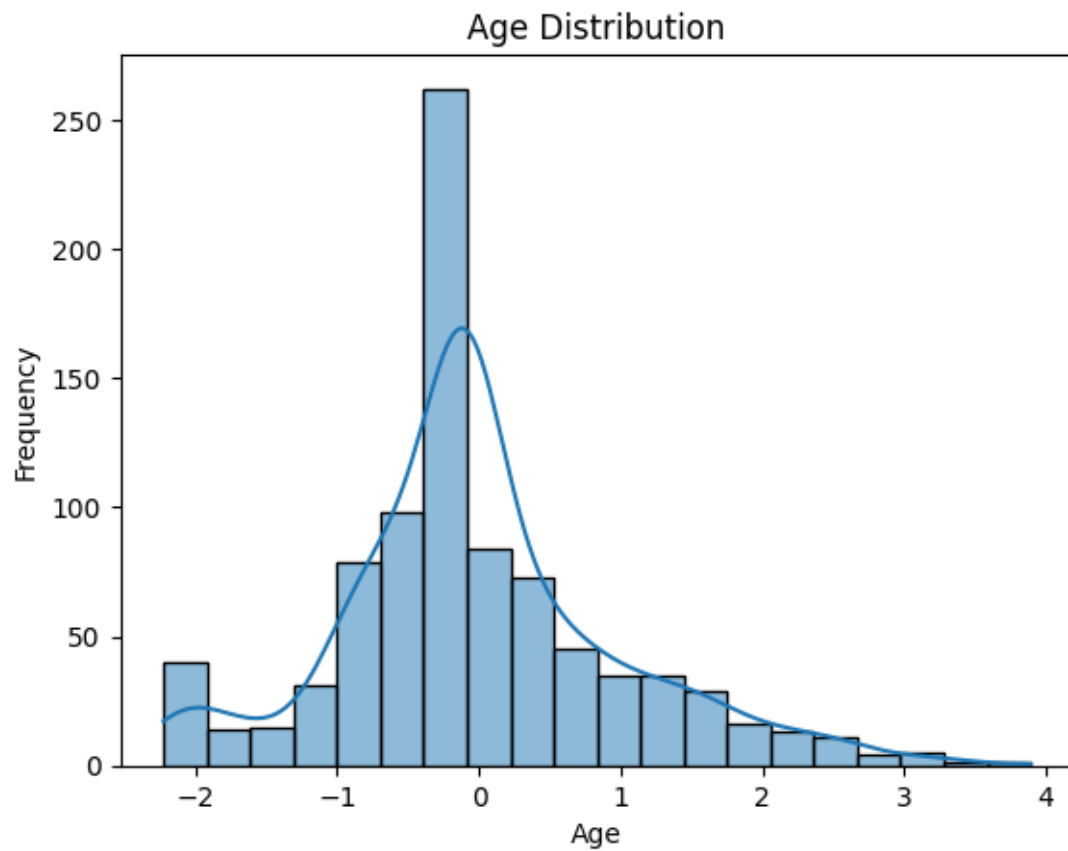
```

```

[12]: sns.histplot(df["Age"], bins=20, kde=True)

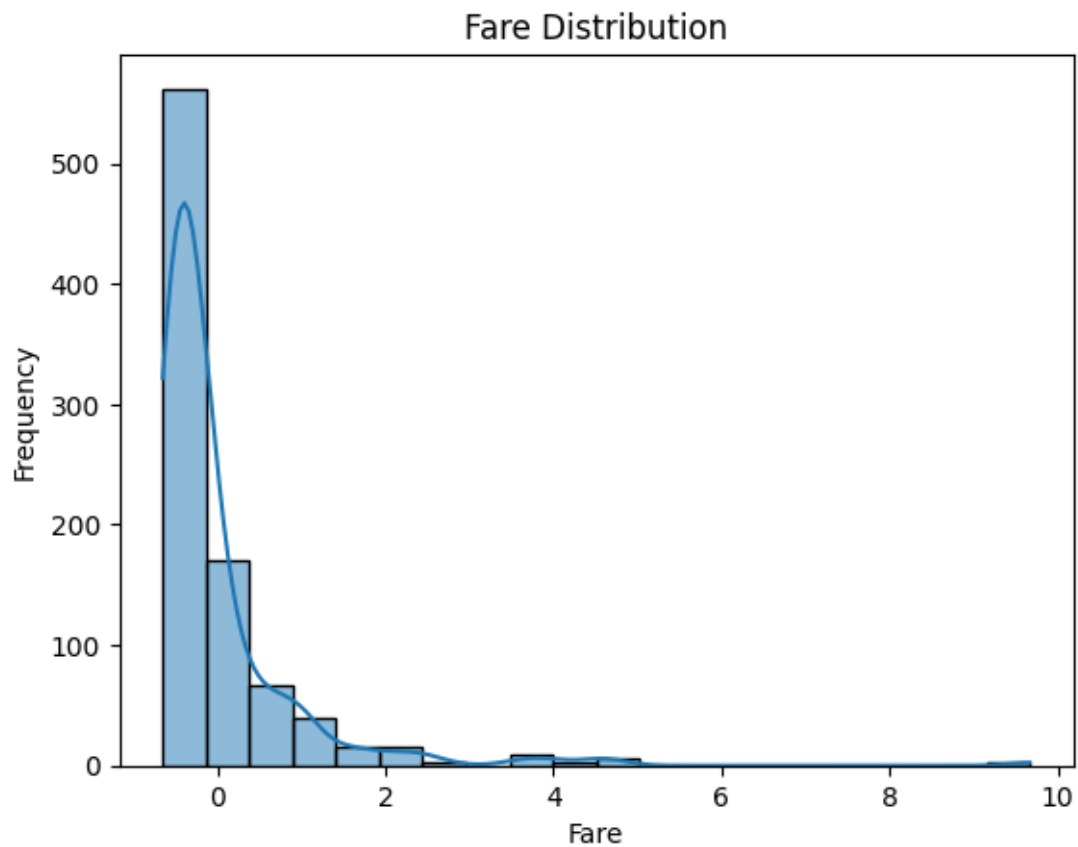
      plt.title("Age Distribution")
      plt.xlabel("Age")
      plt.ylabel("Frequency")
      plt.show()

```

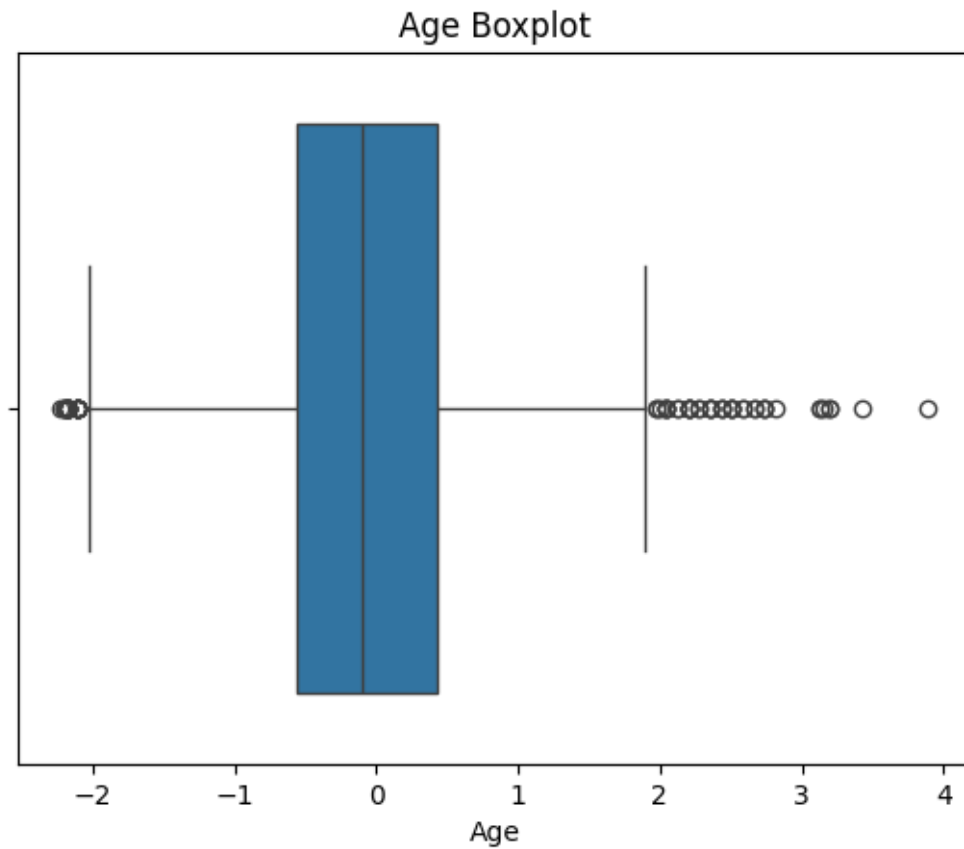


```
[13]: sns.histplot(df["Fare"], bins=20, kde=True)

plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```

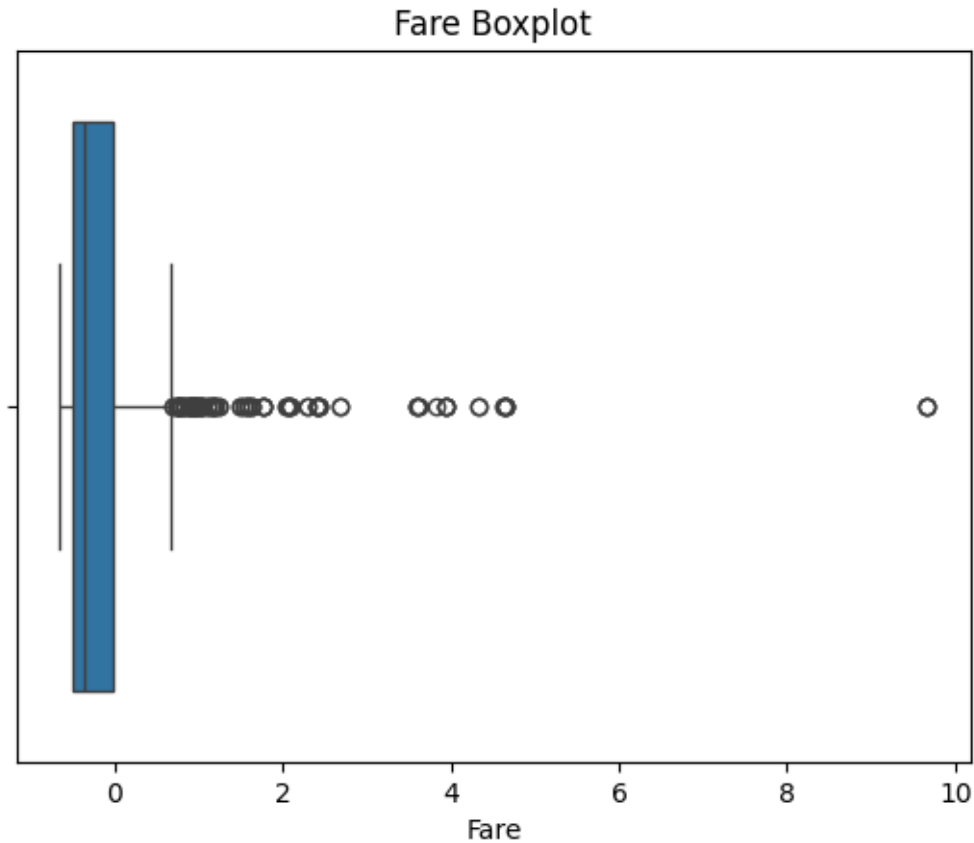


```
[14]: sns.boxplot(x=df["Age"])  
  
plt.title("Age Boxplot")  
plt.xlabel("Age")  
plt.show()
```



```
[15]: sns.boxplot(x=df["Fare"])

plt.title("Fare Boxplot")
plt.xlabel("Fare")
plt.show()
```



```
[16]: features = ["Pclass", "Sex", "Age", "SibSp", "Parch", "Fare", "Embarked"]

X = df[features]
y = df["Survived"]
```

```
[17]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

Training data: (712, 7)

Testing data: (179, 7)

[]: